

Question 3:

Data Dictionary - identity data element, element description from one data element to make a form (pick the data element carefully)

Data Element Description: Customer Number

What is a Data Element?

A data element is a single, uniquely defined piece of information stored in a system. Each element is described once in a data dictionary and includes characteristics such as its name, type, format, validation rules, and usage. Data elements ensure consistency across applications and allow accurate data capture, storage, and reporting.

Chosen Data Element: Customer Number

We have chosen **Customer Number** as the data element because it uniquely identifies each customer in the system. It is widely used across transactions, orders, billing, and reporting, making it critical for system accuracy and integration.

Characteristics of Customer Number:

- **Element ID:** CUST_NUM_001 (unique identifier in the data dictionary)
- **Aliases:** Client Number, Receivable Account Number (other names used in different systems)
- **Description:** Uniquely identifies customers who have conducted business within the last five years; includes a system-generated check digit for validation.
- **Base / Derived:** Derived — generated by the system to ensure uniqueness and prevent manual errors.
- **Length:** 6 digits
- **Data Type:** Numeric (integer)
- **Input Format:** 6-digit number only
- **Output Format:** 6-digit number with check digit
- **Default Value:** None

- **Validation Criteria:** Must pass modulus-11 check; value between 1 and 999999
- **Comments:** Used in all order processing, billing, and customer service systems. Old system aliases must be updated to match this canonical definition.

Element Description Form

ID	
Name	Customer Number
Alias	Client Number
Alias	Receivable Account Number
Description	Uniquely identifies a customer who has made any business transaction within the last five years

Element Characteristics

Length	6	Dec. Pt.	☐ Alphabetic
Input Format	9 (6)		☐ Alphanumeric
Output Format	9 (6)		☐ Date
Default Value			☒ Numeric
☒ Continuous or		☐ Discrete	☐ Base or ☒ Derived

Validation Criteria

Continuous		Discrete	
Upper Limit	<9999999	Value	Meaning
Lower Limit	>0		

Comments The customer number must pass a modulus-11 check digit test. It is derived because it is computer generated and a check digit is added.

Chosen Data Element: Port of Call

We have chosen **Port of Call** because it identifies the specific stop locations on cruise routes. It is critical for booking, itinerary planning, and availability checks.

Characteristics of Port of Call:

- **Element ID:** PORT_CALL_001
- **Aliases:** Destination, Stop Location
- **Description:** Represents a port where a cruise ship will stop. Examples include Apia, Bora Bora, Papeete.
- **Base / Derived:** Base — entered manually or from an authoritative list of ports.
- **Length:** 15 characters (to accommodate the longest port name, e.g., “Christ Church”)
- **Data Type:** Alphanumeric (text)
- **Input Format:** Up to 15 letters (allowing spaces; no numbers or special characters except hyphen if needed)
- **Output Format:** Up to 15 letters
- **Default Value:** None
- **Validation Criteria:** Must match one of the approved ports in the master list
- **Comments:** Used in cruise itineraries, availability inquiries, and booking systems.

Element Description Form

ID PORT_CALL_001

Name Port of call

Alias Destination

Alias Stop location

Description Identifies a port where a cruise ship will stop

Element Characteristics

Length 15 Dec. Pt. N/A ☐ Alphabetic

Input Format X(15) ☒ Alphanumeric

Output Format X(15) ☐ Date

Default Value ☐ Numeric

☐ Continuous or ☒ Discrete ☒ Base or ☐ Derived

Validation Criteria

Continuous	Discrete	
	Value	Meaning
Upper Limit <u> </u>	<u>Apia</u>	<u>Port in Samoa</u>
	<u>Pago Pago</u>	<u>Port in American Samoa</u>
Lower Limit <u> </u>	<u>Moorea</u>	<u>Port in French Polynesia</u>

Comments Used in cruise itineraries, bookings, availability checks

Important to remember -

Data Type	Meaning
Bit	A value of 1 or 0, a true/false value
Char, varchar, text	Any alphanumeric character
Datetime, smalldatetime	Alphanumeric data, several formats
Decimal, numeric	Numeric data that are accurate to the least significant digit; can contain a whole and decimal portion
Float, real	Floating-point values that contain an approximate decimal value
Int, smallint, tinyint	Only integer (whole digit) data
Currency, money, smallmoney	Monetary numbers accurate to four decimal places
Binary, varbinary, image	Binary strings (sound, pictures, video)
Cursor, timestamp, uniqueidentifier	A value that is always unique within a database
Autonumber	A number that is always incremented by one when a record is added to a database table

Formatting Character	Meaning
X	May enter or display/print any character
9	Enter or display only numbers 9(6)
Z	Display leading zeros as spaces
,	Insert commas into a numeric display
.	Insert a period into a numeric display
/	Insert slashes into a numeric display
-	Insert a hyphen into a numeric display
V	Indicate a decimal position (when the decimal point is not included)

Date - MMDDYYYY

Numeric - storage type mention

Decimal - zoned, packed, binary
↓
to save space
← same reason but less used

$X(8) \rightarrow$ xxxxxxxx characters

Alphabetic $\rightarrow$ only letters

* Suppose a numeric element 0072

Format $\rightarrow$ zzzz

Display $\rightarrow$ 72 (easier)

* 1234567

$\rightarrow$ 9,999,999

Display $\rightarrow$ 1,234,567

* 124.5

$\rightarrow$ 999.9

* 51234.5

$\rightarrow$ 99,999.9

51,234.5

* 999V99 $\rightarrow$ 123.45

* MM/DD/YY (date)

250923

$\rightarrow$ 25/09/23

* 99/999

12345

$\rightarrow$ 12/345

fraction

Base vs Derived



entered
directly by
users/
captured
from external
source

$\swarrow$ calculated/generated from base
elements/derived elements

Validation criteria —

- Range (upper limit — lower limit)
- Value list (discrete val)
- check digit / Formula (e.g. Modulus 11 check)
- Format
- Length & decimal pts.